

Assignment 6
40 points

Stochastic Calculus
for Finance

Due Sunday, May 3

The assignment has to be handwritten (written on paper or an iPad or another writing tablet). The assignment submission should contain the following:

1. The discussion on interest rate dependent assets
 - a) Interest rate process and discount process
 - b) Definition and risk-neutral pricing of zero coupon bonds, coupon paying bonds
 - c) Definition of forward contracts on an asset, m -forward price of an asset, m -forward interest rates
 - d) Definition and risk-neutral pricing of interest rate contracts, m -period interest rate swaps, caps, floors, caplets and floorlets
2. Exercise 6.3 (The risk neutral pricing of an interest rate contract) from Shreve, Part 1
3. Exercise 6.4 (The hedging portfolio for a interest rate caplet) from Shreve, Part 1